

08 Video as World Simulator

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10 MIN READ · INTERACTIVE

Add one input to a video model and it stops being a video.
What scaling that bought, and why matching physics
across the range you trained on is not the same as having
the rule.

Take a model that predicts the next frame of video, and add one input.

Alongside the frames it has already seen, tell it which key is being held. Now the
same opening gives you one continuation if you hold left and a different one if you
hold right. It has stopped being a video and started being somewhere you are.

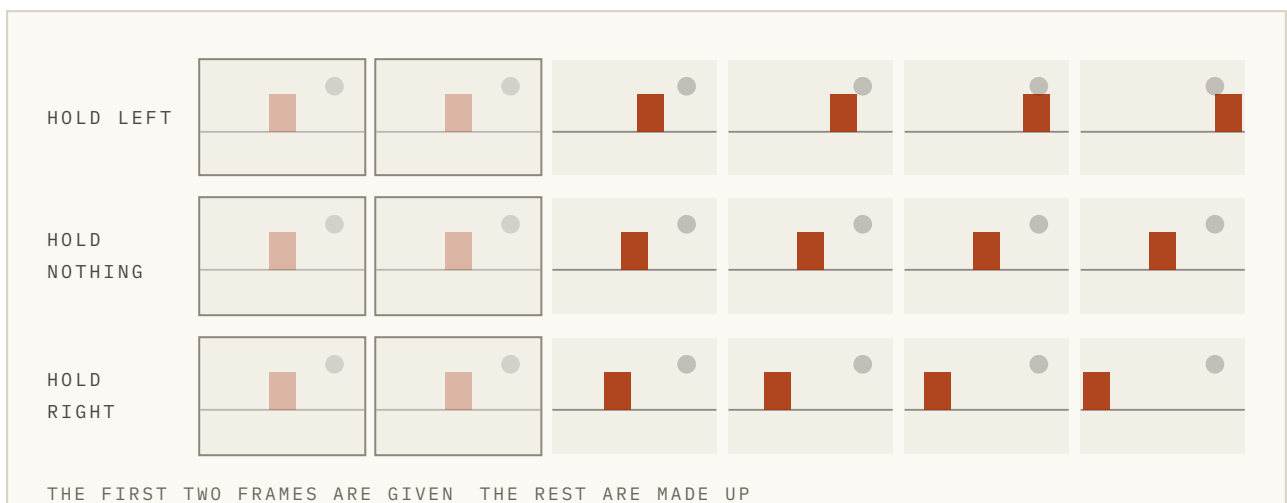


FIG. 8.1 One model, one start, three futures. The first two frames are given; the rest are made up. Turn the action input off and all three rows become identical, which is the whole difference: without it there is exactly one future on offer, and it is whichever one the model finds most likely.

That is the entire architectural move. Everything else in this chapter is what happens when you do it at scale.

What scaling bought

More than most people expected, including most of the people building it.

Genie 3 is reported to generate interactive video at 720p and 24 frames a second. It is reported to hold a scene together over several minutes, and to take instructions part-way through: change the weather, add a flock of birds. Walk away from something, walk back, and it is roughly as you left it.

Take those numbers as the lab's report rather than as measurements anyone outside it has repeated. That is not a slight; it is the normal state of this subject, and Chapter 9 is largely about how little of it can currently be checked.

Nothing in that system stores the scene. There is no geometry behind the picture, no list of objects, nothing you could export. The room stays put because the model got extremely good at continuing what it had already drawn, and continuing consistently is part of continuing well.

Which is a stranger achievement than it sounds. A system with no memory of a room, reconstructing that room from its own previous output, well enough that you cannot catch it out in a minute of looking.

What "emergent physics" means

You will see the claim made about every one of these systems. It is worth being precise about what it can and cannot mean.

The observation behind it is real. Train on enough video and the outputs start respecting things nobody put in. Dropped objects accelerate downwards. Water pools rather than piling up. Things hidden behind other things come back when the thing in front moves. No equation was supplied. Those regularities are in the training footage, and predicting the footage well requires reproducing them.

The claim built on top is where the trouble starts. "It learned physics" suggests it acquired the rule, and there is a test for that. It is not whether the outputs look right on the sort of thing it was trained on.

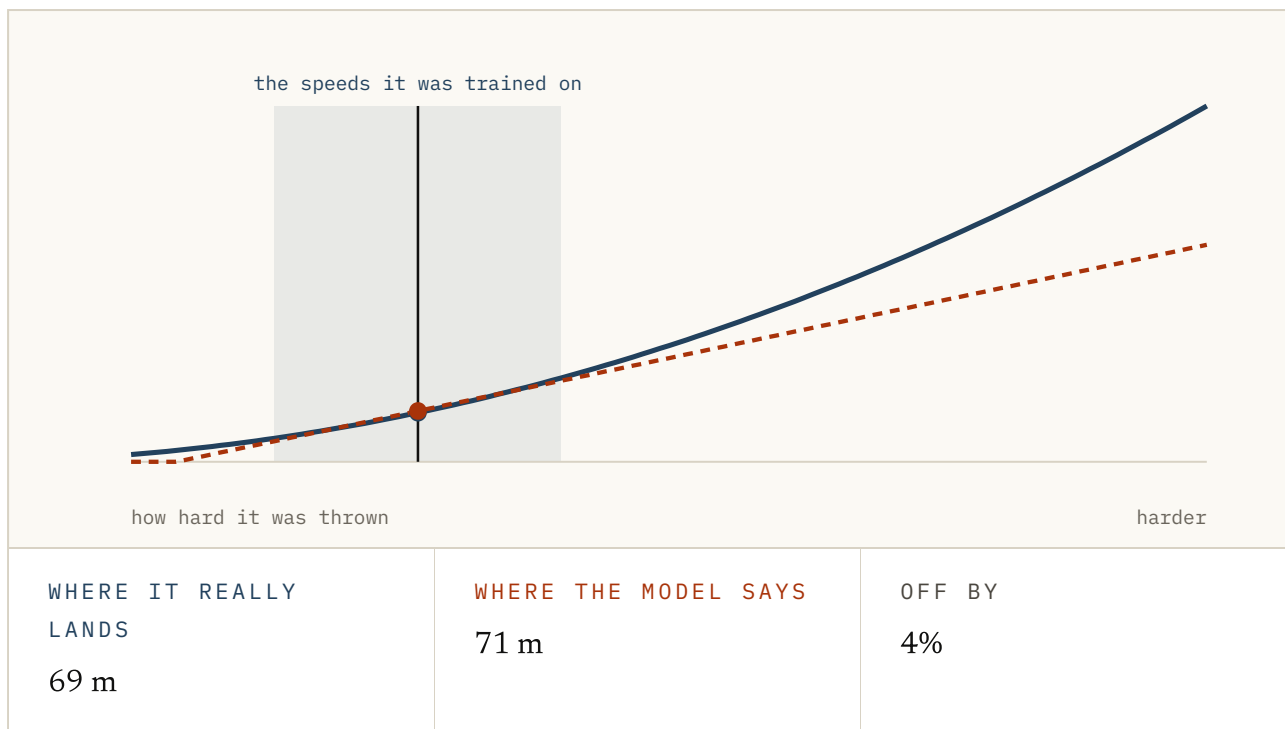


FIG. 8.2 The dashed line is a model fitted to the true behaviour across one band of speeds. Inside that band it is within a few per cent, and anyone testing it there would conclude it had the rule. Drag outside the band. It does not have the rule; it has a straight line through the part of the curve it was shown.

Everything inside the band is consistent with having learned physics. It is equally consistent with having learned the answers to the questions it was asked. The two only come apart outside, and outside is exactly where nobody tests, because outside is where you have no footage to compare against.

So the honest version of the claim is narrower and more useful. These systems reproduce a great many physical regularities across the range of situations they were trained on. Whether that constitutes having the rule is not settled by any demonstration you have seen, because the demonstrations are all inside the band.

This is not a special failing of video models. It is the ordinary problem with any fitted function, and the reason it feels different here is only that the outputs are so persuasive. A wrong answer that looks like a photograph is more convincing than a wrong number.

What they are for

Given all that, it would be easy to be dismissive, and that would be a mistake.

For training other systems, a generated world is very useful. It costs nothing to run. It resets instantly. It will produce a thousand variations of a scene that would take a week to stage. NVIDIA's Cosmos is built and marketed on exactly this: physically-aware generated video as training data for robots and vehicles.

For being a world you can move around in, they are already good. That is worth something on its own, and no other approach in this course delivers it.

For being the thing an agent plans inside, they are the wrong shape. A planner needs to try a hundred action sequences and score them, which means running the model a hundred times, and running these models is expensive. It also needs a compact state to reason over, and a frame is the opposite of compact.

Where the seams are

Three, briefly, because the next chapter is the full accounting.

Consistency is learned, not enforced. Nothing prevents the room from changing. It usually does not, and the fact that it usually does not is a statistical property of a trained system rather than a guarantee.

The interesting moments are the vague ones. A model trained on pixels is pushed towards the average of the possible futures wherever the future is open. The future is most open exactly when something is about to happen.

Nobody agrees how to score them. There is no accepted measurement of whether a generated world is behaving, which means claims about improvement are mostly claims about how the outputs look to the person making them.

Try it

1 What single change turns a video model into something you can steer?

- a. More parameters
- b. Conditioning each generated frame on an action as well as on the frames before it
- c. Higher resolution
- d. A longer context window

ANSWER b. Conditioning each generated frame on an action as well as on the frames before it. Without an action input there is exactly one future on offer: whichever the model finds most likely. With it, the same start has as many continuations as there are things you could do.

2 A generated room stays as you left it when you turn back. What is holding it there?

- a. Stored geometry behind the picture
- b. A list of objects in memory
- c. Nothing. The model got good at continuing consistently, which is part of continuing well
- d. A separate persistence module

ANSWER c. Nothing. The model got good at continuing consistently, which is part of continuing well. There is no scene behind the frames. Consistency is a statistical property of a well-trained system, which is a stranger achievement than storing it would be.

3 Dropped objects in generated video accelerate downwards, and nobody supplied gravity. What does that establish?

- a. The model has learned the law of gravity
- b. That the regularity is in the training footage, and reproducing it is required to predict the footage well
- c. The model contains a physics engine
- d. Nothing at all

ANSWER b. That the regularity is in the training footage, and reproducing it is required to predict the footage well. The observation is real. The question is whether reproducing a regularity across the range you trained on amounts to having the rule, and the observation alone does not settle it.

4 In Figure 8.2, the model is within a few per cent across the band it was trained on. Why is that not evidence it has the rule?

- a. The measurement is unreliable
- b. Matching inside the band is equally consistent with having learned the answers to the questions it was asked
- c. A few per cent is too large an error
- d. The band was too narrow

ANSWER b. Matching inside the band is equally consistent with having learned the answers to the questions it was asked. Having the rule and having a fit through the sampled region only come apart outside the band, and outside is where nobody tests because there is no footage to compare against.

5 Why is a generated video world the wrong shape for an agent to plan inside?

- a. It is not accurate enough
- b. Planning means running the model many times over, and a frame is the opposite of a compact state
- c. It cannot be conditioned on actions
- d. It has no reward signal

ANSWER b. Planning means running the model many times over, and a frame is the opposite of a compact state. A planner needs to try many action sequences and score them cheaply. These models are expensive to run and produce frames rather than something small to reason over.

6 What are generated worlds unambiguously good for right now?

- a. Replacing physics engines
- b. Producing training data and being a place you can move around in
- c. Long-horizon planning
- d. Compressing video

ANSWER b. Producing training data and being a place you can move around in. They reset instantly, cost nothing to run, and produce endless variation. And no other approach in this course gives you an actual picture to walk into.

7 Why does a wrong answer from one of these systems feel more convincing than a wrong number?

- a. The errors are smaller
- b. It arrives as something that looks like a photograph
- c. The models are better calibrated
- d. It usually is not wrong

ANSWER b. It arrives as something that looks like a photograph. The extrapolation problem here is the ordinary problem with any fitted function. What is different is how persuasive the output is when it fails.

8 Claims that these systems are improving are hard to check. Why?

- a. The models are proprietary
- b. There is no accepted measurement of whether a generated world is behaving
- c. The improvements are too small
- d. Nobody publishes results

ANSWER b. There is no accepted measurement of whether a generated world is behaving. Without an agreed score, most claims about improvement are claims about how the outputs look to the person making them.

FIG. 8.3 Eight questions on the move and the claim. The middle three are the ones worth being careful about when a demo is impressive.

Which brings us to the part of the subject that is usually left out of the announcements.

Thanks for reading. Chapter 9 is the honest accounting: what drifts, what gets forgotten, what cannot be asked, and why nobody can currently tell you which of these systems is best.

Sources

Genie: Generative Interactive Environments BRUCE ET AL., 2024 ↗

Latent actions learned from unlabelled video, which is the move that turns a video model into somewhere you can be.

<https://arxiv.org/abs/2402.15391>

Genie 3 GOOGLE DEEPMIND, 2025 ↗

The reported numbers this chapter quotes: 720p, 24 frames a second, minutes of coherence. A lab report rather than a result anyone outside has repeated.

<https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>

Diffusion Models Are Real-Time Game Engines (GameNGen) VALEVSKI ET AL., 2024 ↗

DOOM generated frame by frame from previous frames and inputs, fast enough to play. The clearest demonstration that this is playable rather than merely watchable.

<https://arxiv.org/abs/2408.14837>

How Far is Video Generation from World Model: A Physical Law Perspective

KANG ET AL., 2024 ↗

The measured version of Figure 8.2. Video models match physical laws within the distribution they were trained on, and do not extrapolate outside it.

<https://arxiv.org/abs/2411.02385>

Video generation models as world simulators OPENAI, 2024 ↗

The report that made the emergent-physics claim a mainstream one. Worth reading for exactly what it does and does not assert.

<https://openai.com/index/video-generation-models-as-world-simulators/>

Cosmos World Foundation Model Platform for Physical AI NVIDIA, 2025 ↗

Generated video built as training data for robots and vehicles, which is the use these systems are unambiguously good for.

<https://arxiv.org/abs/2501.03575>

Cosmos NVIDIA ↗

The product framing, and a useful boundary case: generative video beside explicit simulation, sold as one platform.

<https://www.nvidia.com/en-us/ai/cosmos/>

FIG. 8.4 Ordered for someone building on this rather than for historical completeness.